Simultaneous determination of ascorbic acid and uric acid using Cr - SnO_2 nanoparticles modified glassy carbon electrode

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***Abstract*— An electrochemical sensor for simultaneous determination of ascorbic acid and uric acid have been fabricated using Cr doped SnO_2 nanoparticles (NPs) as sensing material. The Cr-SnO_2 NPs (Cr: 0 and 1wt %) have been synthesized by microwave irradiation method. Cr doping into SnO_2 enabled the particle size reduction and improvement in the sensing characteristics of the semiconducting compound SnO_2. 1wt% Cr-SnO_2 NPs modified glassy carbon electrode (GCE) exhibits excellent electrocatalytic activity towards the simultaneous detection of ascorbic acid (AsA) and uric acid (UA) with enhanced oxidation current when compared to that of bare and SnO_2/GCE. The Square wave voltammetry data showed that the obtained anodic peak currents linearly dependent on the AsA and UA concentration in the range of 0.1×10^{-6} to 0.5×10^{-3} M and 0.1×10^{-6} to 62×10^{-6} M respectively with the detection limit of 36×10^{-9} M and 14×10^{-9} M under optimum conditions. The fabricated sensor has several advantages like easy preparation, high sensitivity, stability and good reproducibility towards the AsA and UA determination. Also, the modified electrode exhibited good anti-interferent activity for several potential interferents including dopamine (DA). Determination of AsA and UA concentration in pharmaceutical products, fruits, urine and blood serum by the proposed sensor showed satisfactory result.**

Keywords: Ascorbic acid, uric acid, square wave voltammetry, Cr doped SnO_2 nanoparticles

I. INTRODUCTION

Nanostructured metal oxides (NMOs) have attracted considerable attention in chemical and biosensor technology due to their large variety of physical, electrical and molecular properties such as size, shape, effective surface area, functionality, electron transfer, adsorption capability and assembly properties [1]. Various synthesis procedures are adopted for preparing NMOs with well-controlled morphological and structural properties. In addition, the introduction of chemical dopants offers a convenient method for producing functional materials. It has been demonstrated that the properties of metal oxides can be enhanced and tuned through the addition of various dopants and processing methods [2]. Among various metal oxides, tin dioxide (SnO_2), a stable n-type semiconductor (Eg = 3.6 eV at 300 K), has been extensively studied as a promising material for a variety of applications including gas sensors, catalysis, transparent conducting coatings for furnaces and electrodes, Li-ion batteries, solar cells etc [3]. The SnO_2 is also a good candidate for biosensor applications due to high isoelectric point (IEP ~5.0), non-toxicity, high catalytic efficiency, strong adsorption ability, good biocompatibility and high electron communication features [4]. Recently, the research interest has been extended to modify SnO_2 nanostructures through doping with transition metals such as Co, Mn, Fe, Ni, and Cr etc which improves the electrochemical properties and synergistic effects of SnO_2 [5]. Among these, Cr is an important anti-ferromagnetic transition element with abundant electron cell structure, and its ionic radius 62 pm (Cr^{3+}) is close to that of Sn^{4+} (74 pm), which means Cr can easily be substituted at Sn^{4+} site in the SnO_2 crystal lattice. We have synthesized Cr doped SnO_2 NPs (Cr content: 0, 1 & 3 wt%) by microwave irradiation method. It is found that the magnetic behaviour of SnO_2 NPs changes from diamagnetic (0 & 1wt% Cr) to ferromagnetic (3 wt% Cr) depending upon the dopant concentration. In a previous work, we have shown that the 3 wt% Cr doped SnO_2 NPs are highly suitable as sensing element for the detection of riboflavin (vitamin B_2) with improved sensitivity and selectivity [6]. Here, we demonstrate that the diamagnetic 1wt% Cr-SnO_2 NPs is highly useful for the simultaneous detection of ascorbic acid and uric acid.

L-Ascorbic acid and uric acid are crucial biomolecules for physiological processes in human metabolism and usually coexist in biological samples (human blood and urine). AsA is one of the major water soluble antioxidant within the body. It readily donates electrons to break the chain reaction of lipid peroxidation. Its water soluble properties allow quenching of free radicals before they reach the cellular membrane [7]. In addition,

AsA is extensively used for the prevention and treatment of the common cold, some mental illnesses, gingivitis, skin diseases, diarrhoea, malaria and bacterial infections. It has been a necessary nutrient for repairing and maintaining cells and bones as well as fighting infection and lowering both cancer and cardiovascular diseases risks [8]. AsA deficiency results in the scurvy, immunity deficiency and wound healing, anticancer and antitumor activity. Uric acid, the final product of purine metabolism, has been found to be associated with many diseases such as gout, cardiovascular disease, hypertension and obesity. High concentrations of UA in blood cause deposition of urate crystals, which could ultimately result in chronic joint inflammation, renal impairment and high risk factors for atherosclerosis [9]. It is well known that the dopamine coexists in the extracellular fluid of the central nervous system and serum, and so simultaneous detection of DA, AsA and UA by electrochemical methods is being attempted. Though these biomolecules are electrochemically active, their simultaneous detection by unmodified electrodes is very difficult due to the fact that they have similar oxidation potentials at most solid electrodes, preventing selective identification.

Various modified electrodes including carbon nanofibers [10], Fe_3O_4@Au nanoparticles [11], nitrogen doped graphene [12], pyrolytic graphite [13], poly(N-methylpyrrole)/Pd-nanoclusters [14] and SnO_2/MWCNT [17] have been used for the simultaneous determination of AsA and UA. Here we report the simultaneous detection of ascorbic acid and uric acid sensor using 1wt% Cr doped SnO_2 NPs in the presence of DA. The fabricated sensor exhibits an excellent sensitivity, wide linear range, good reproducibility, high stability and the lowest detection limit of 36 nM for AsA and 14 nM for UA respectively. The analytical performance of the sensor has been evaluated in pharmaceutical products, fruits, human urine and blood samples with satisfactory results.

II. EXPERIMENTAL

2.1 Materials

The reagents used were of analytical grade without further purification; $Cr(NO_3)_3.9H_2O$ and $SnCl_2.2H_2O$ (Fischer Scientific, Mumbai) and ascorbic acid (Sisco Research Laboratories, Mumbai). Stock standard solution of ascorbic acid and uric acid was prepared by dissolving appropriate amounts of AsA and UA in 20 mL of 0.1 M phosphate buffer solution (PBS, pH 7.0), and stored at 4°C. The PBS was prepared by mixing appropriate amounts of Na_2HPO_4 and NaH_2PO_4. The solutions were prepared with double-distilled water.

2.2 Measurements and apparatus

Cyclic voltametric and amperometric measurements were performed on CHI 608d electrochemical workstation (CH Instruments, Austin, USA). A conventional two compartment three electrode cell with a mirror polished 3 mm glassy carbon (GC) as the working electrode, Pt wire as counter electrode and a Ag/AgCl (3M KCl) as reference electrode. All experimental solutions were deoxygenated by bubbling highly pure nitrogen and maintained under nitrogen atmosphere during measurements. Cyclic voltammograms were recorded between a potential window of -0.2 V and 0.8 V in 0.1 M PBS (pH 7.0) containing various concentrations of ascorbic acid and uric acid. Amperometric experiments were carried out in an electrochemical cell containing 10 mL of 0.1 M phosphate buffer at pH 7.0 under constant stirring. A magnetic stirrer (270 rpm) and a stirring bar provided the convective transport in the amperometric experiments. Square wave voltammograms (SWVs) were obtained by scanning potential from −0.2 to 0.8 V with frequency of 10 Hz, amplitude of 50 mV and step potential of 5 mV.

Surface morphology and particle size analyses were done using transmission electron microscopy (JEOL 2100 F) with an accelerating voltage of 200 keV.

2.3 Preparation and analysis of real samples

Pharmaceutical products, fruits and human urine, blood serum were selected for AsA abd UA real sample analysis. A kind of B complex forte with vitamin C, B complex with vitamin C (Pfizer Ltd, Mumbai) and Biotin and vitamin C tablet (Glaxosmithkine Pharmaceutical Ltd, Mumbai) were obtained from a local drugstore. Human urine and blood serum samples were collected from Government Hospital, Karaikudi. The tablets were weighed, and ground into fine powder with a mortar and a pistol. Each product was mixed with 100 mL of deionized water, sonicated for 30 minutes and then filtered through Whatman 42 (125 mm diameter) filter paper which yielded a clear solution. The solution was diluted with phosphate buffer (pH 7.0) to the required concentration and amperometry was performed with standard addition method. Various fruits like orange, lemon, mango and tomato were chosen for the analysis of AsA. Fruit juices were prepared independently by a mechanical press and then centrifuged in order to prepare clear solution. Further purification was done through filtration with Whatman 42 (125 mm diameter) filter paper. Then, 500 µL of sample solution was added with PBS and analyzes were done as described in section 3.6.

2.4 Preparation of SnO_2 NPs

SnO_2 NPs were synthesized by microwave irradiation method as reported earlier [5]. Briefly,

0.1 M solution of $SnCl_2.2H_2O$ and $Cr(NO_3)_3.9H_2O$ with Cr content of (1wt%) were mixed. Ammonium hydroxide (NH_4OH) was added drop wise to the above solution until pH reaches 8. This solution was again stirred for 30 minutes to form a greenish colloidal gel. Further, the hydroxide product (Sn_4OH) was washed several times with deionised water and ethanol in order to remove the Cl^- and NH^{4+} ions respectively. Then the precipitate was transferred into microwave oven (600 W) and kept for 20 minutes. Finally, the Cr doped SnO_2 NPs were obtained after annealing the precipitates at 600 °C for 5 hrs in ambient atmosphere.

2.5 Preparation of Cr-SnO_2 modified GC electrode

Prior to modification, the GCE was sequentially polished with 1.0, 0.3 and 0.5 µM gamma alumina slurry respectively, and rinsed thoroughly with distilled water after each polishing step. After that, the electrode was successively sonicated in ethanol and water for 2 minutes each, and then allowed to dry at room temperature. The Cr-SnO_2 modified electrode was prepared by a simple drop casting method. 5 mg of Cr-SnO_2 NPs were dispersed in 1 mL double distilled water, and its suspension was obtained with a good dispersion after ultrasonicating for 10 minutes. About 10 µL of suspension was drop casted onto GC electrode and air dried at room temperature for 60 minutes.

III. RESULTS AND DISCUSSION

3.1 Structural characterizations of the SnO_2 NPs

Powder X-ray diffraction (XRD) studies confirmed the phase formation (tetragonal phase; rutile structure) and high purity of pristine SnO_2 and Cr doped SnO_2 NPs [6]. Transmission electron microscopy (TEM) studies revealed that the synthesized NPs have spherical morphology with average particle sizes of 21 and 17 nm for SnO_2 and 1wt% Cr-SnO_2 samples respectively. Figure1 (A, B, C) show a typical TEM and HRTEM pictures of 1wt% Cr-SnO_2 and pure SnO_2 nanparticles respectively. It can be noticed that the Cr doped SnO_2 sample has a twin-like lattice defect (Figure 1B) while the undoped SnO_2 lattice structure was nearly defect free (Figure 1C).

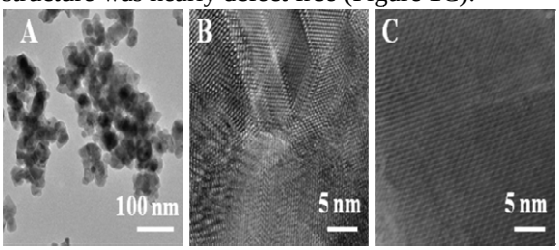

Fig.1 (A) TEM image of 1wt% Cr-SnO_2 NPs and HRTEM image of (B) 1wt% Cr-SnO_2 NPs and (C) pure SnO_2 NPs

3.2 Electrocatalytic activity of ascorbic acid and uric acid on the 1wt% Cr-SnO_2/GCE

The electrocatalytic activity of 1wt% Cr-SnO_2 modified GC electrode toward ascorbic acid and uric acid was investigated by cyclic voltammetry. The CVs of ascorbic acid at bare GC electrode (curve a), undoped SnO_2 (curve b) and 1wt% Cr-SnO_2 (curve c) modified GC electrodes recorded in 1 mM AsA with phosphate buffer solution (pH 7.0) are shown in Figure 2A. The 1wt% Cr-SnO_2 modified electrode shows an oxidation peak at 0.174 V, which was about 120 mV less than that at the bare GC and undoped SnO_2 modified electrodes. The peak current (i_{pa}) value of 1wt% Cr-SnO_2 (1.95 × 10^{-5} A) is much higher when compared to the values for bare GC (1.77 × 10^{-5} A) and undoped SnO_2 (1.71 × 10^{-5} A) modified GC electrodes. The CVs of UA were also investigated independently with100 µM UA using the same set of modified electrodes (Figure 2B). A large anodic peak current was observed at 0.34 V (curve c), which could be attributed to the irreversible reaction of UA oxidation process. Only a very weak current response can be seen at the undoped SnO_2/GCE (curve b) and bare GCE (curve a). The data suggest that the 1wt% Cr doped SnO_2 has good electrocatalytic performance toward AsA and UA sensing when compared to that of the bare GC and undoped SnO_2 modified electrodes. Such a large increase in catalytic activity towards the oxidation of AsA and UA could be attributed to large effective surface area and high electron transfer rate.

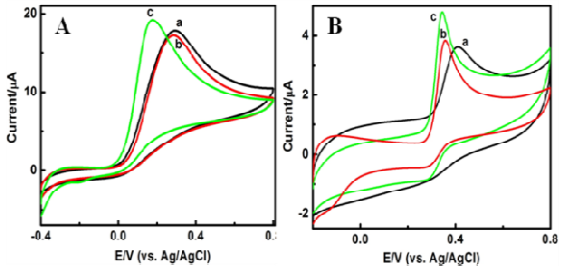

Fig.2 (CVs obtained for (A) 1 mM AsA and (B) 100 µM UA at the (a) bare, (b) undoped SnO_2 and (c) 1wt% Cr-SnO_2 modified GC electrodes in PB (pH 7.0) solution with the scan rate of 50 mV/s.

3.2.1 Simultaneous detection of ascorbic acid and uric acid

The voltammetric behaviors of 500 µM AsA and UA mixture were investigated by CV. As shown in Figure 3, AsA and UA yielded two well-defined oxidation peaks at 0.17 V and 0.34 V at the 1wt% Cr-SnO_2/GCE (curve c) under identical conditions. The voltammetric peak current at 1wt% Cr-SnO_2/GCE is almost double than that of the bare electrode. The enhancements in peak current and difference in oxidation potential could be attributed to the large effective surface area and improved

electrocatalytic properties of the 1wt% Cr-SnO_2 modified GCE toward the oxidation of AsA and UA.

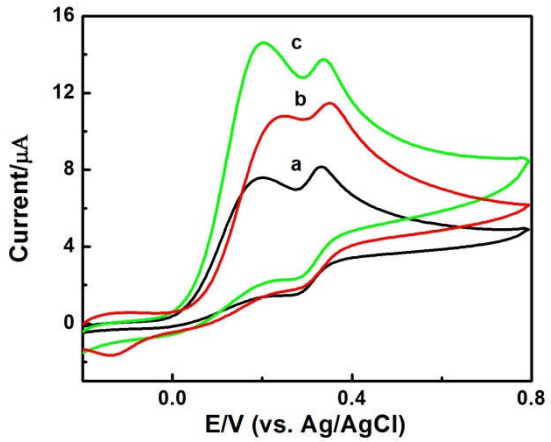

Fig.3 Simultaneous CVs obtained for 500 μM AsA and UA at the (a) bare, (b) undoped SnO_2 and (c) 1wt% Cr-SnO_2 modified electrodes recorded in PB (pH 7.0) solution with the scan rate of 50 mV/s.

The effect of scan rate on the oxidative potential (Epa) and peak current (Ipa) of AsA and UA (Figure 4A & B) at 1wt% Cr-SnO_2/GCE was studied in the range of 0.05–0.5 V s^{-1} in order to investigate the kinetics of electrode reactions. The results showed that the peak currents vary linearly with the scan rate (insets of Figure 4) for both drugs, which confirms a diffusion controlled process for electrochemical oxidation of AsA and UA on the surface of 1wt% Cr-SnO_2/GCE. The oxidation potential shifted to positive values with increasing the scan rate, suggesting the electron transfer is quasi-reversible.

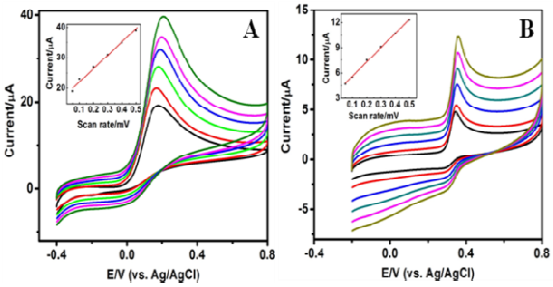

Fig.4 CVs obtained for (A) 1 mM AsA and (B) 100 μM UA at 1wt% Cr-SnO_2/GCE in pH 7.0 PBS at different scan rates of 50-500 mV/s. Inset: plot of anodic peak current vs. scan rate.

Linear sweep voltammograms (LSVs) obtained for AsA and UA in the concentration range of 0-1000 μM and 0 – 100 μM respectively at the nano-Cr-SnO_2 modified electrode is shown in Figure 5 (A & B). The oxidation current of both drugs increases upon each increment of AsA and UA in steps of 100 μM and 10 μM respectively. The oxidation currents have a linear relationship with the concentration of AsA and UA with a correlation coefficient of 0.997 and 0.998 respectively (Inset Figure 5).

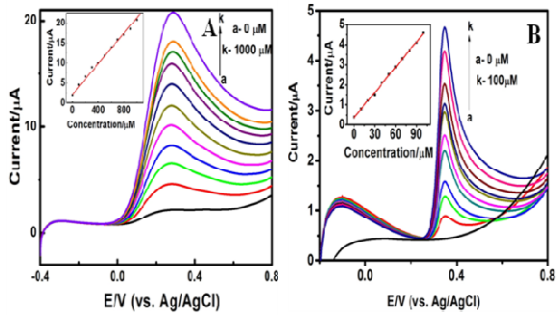

Fig.5 LSVs obtained for (A) AsA and (B) UA in the concentration ranging from 0 to 1000 μM and 0 to 100 μM respectively at the nano-Cr-SnO_2 modified electrode in PB (pH 7.0) solution.

3.3 Amperometric and square wave voltammetric response of the fabricated sensor

The amperometric responses of the 1wt% Cr-SnO_2/GCE with successive additions of ascorbic acid and uric acid to PBS (pH 7.0) at an applied potential of 0.17 V for AsA and 0.34 V for UA are shown in Figure 6 (A&B). Upon addition of both drugs to the PBS, a stepwise increase of oxidation current was observed. The amperometric current has a good linear relationship with concentration of AsA and UA in the range of 0.5×10^{-6} to 0.7×10^{-3} M (R^2=0.999) and 0.5×10^{-6} to 5×10^{-5} M (R^2=0.997) respectively. The corresponding calibration curve of the sensor is shown in the inset of Figure 6. From this slope, the detection limit of AsA and UA was estimated as 47 $\times10^{-8}$ M and 32 $\times10^{-9}$ M respectively. In both the cases, the 1wt% Cr-SnO_2/GCE reached 95% of the steady-state current within 5 s, indicating a rapid electrocatalytic response.

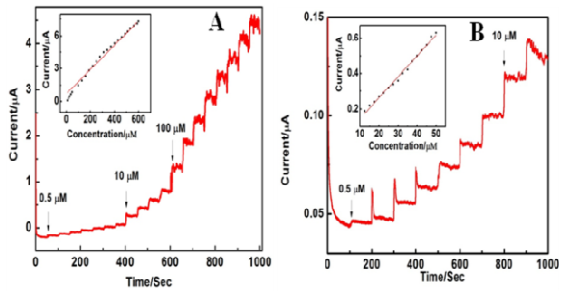

Fig.6 Amperometric response of the 1wt% Cr-SnO_2/GCE with successive addition of (A) AsA and (B) UA into 0.1 M PB solution (pH 7.0) measured at 0.17 V and 0.34 V. Inset: the calibration curve of the sensor.NPs

Subsequently, square wave voltammetry (SWV) method was used to obtain a better resolution of the lowest detection limits. For individual determination of AsA and UA at the 1wt% Cr-SnO_2/GCE, SWV was carried out in the potential

range of -0.2 to 0.8 V in 0.1 M PBS under the frequency of 10 Hz and amplitude of 50 mV. Under these optimum conditions, the electrochemical response of AsA and UA increases linearly with the increase of the target biomolecule concentration (Figure 7). The linear ranges for the determination of AsA and UA are found to be 0.1×10^{-6} to 0.5×10^{-3} M (R^2 = 0.997) and 0.1×10^{-6} to 6.2×10^{-5} M (R^2 = 0.998) with the detection limit of 36 x 10^{-9} M and 14 x 10^{-9} M respectively. The analytical performances such as linear range and the lowest detection limit of the proposed sensor were compared with other simultaneous determination of these biomolecules reported in the literature (Table 1).

TABLE 1 : COMPARISON OF THE RESPONSES OF OTHER ASCORBIC ACID AND URIC ACID SENSORS

Electrode	Method	Linear range (μM)		Detection limit (μM)		References
		AsA	UA	AsA	UA	
[a]CNF/CPE	DPV	2–64	0.8–16.8	2	0.2	[10]
[b]Fe_3O_4@Au–S–Fc/GS-CS/GCE	DPV	6–350	1–90	1	0.2	[11]
[c]NG/GCE	DPV	5 - 1300	0.1 - 200	2.2	0.45	[12]
[d]PGE	DPV	25 - 500	2.5–30	13	1.4	[13]
[e]PMPy/ Pt	DPV	50 - 1000	0.5 to 20	7	0.027	[14]
[f]MCM-41/CPE	DPV	40–400	0.5–75	3	10	[15]
[g]N-PCNPs/GCE	DPV	80–2000	4–50	0.740	0.021	[16]
[h]SnO_2/MWCNT/CPE	DPV	100 - 5000	3-200	50	1	[17]
[i]1wt% Cr-SnO_2/GCE	SWV	0.1 - 500	0.1–62	0.036	0.014	Present work

[a] Carbon nanofibers modified carbon paste electrode
[b]Fe_3O_4@Au-S-Fe/GS-Chitosan modified glassy carbon electrode
[c]Nitrogen doped graphene modified glassy carbon electrode
[d]Pyrolytic graphite activated electrode
[e]Polyn(N-methylpyrrole) Pd nanocluster modified glassy carbon electrode
[f]Mesoporous silica modified carbon paste electrode
[g]Nitrogen doped porous carbon nanopolyhedra modified glassy carbon electrode
[h]SnO_2/multiwalled carbon nanotube modified carbon paste electrode
[i]Cr doped SnO_2 NPs modified glassy carbon electrode

This detection limit is notably lower than the results obtained at similar sensors based on pyrolytic graphite electrode, polyn(N-methylpyrrole) Pd nanocluster mesoporous silica and SnO_2/MWCNT modified electrodes [13-17].The results suggest that the sensor fabricated in the present work can potentially be used for monitoring the concentration of AsA and UA sensitively. Thus, the proposed biosensor (1wt% Cr-SnO_2/GC electrode) exhibits excellent performance in terms of wide linear range and low limit of detection for the simultaneous determination of AsA and UA. In addition, the fabrication process of the present modified electrode is simple, less time consuming, highly stable and reproducible when compared to the other methods reported in literature.

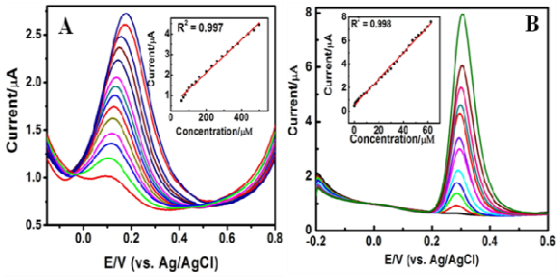

Fig.7 Square wave voltammograms recorded for 1wt% Cr-SnO_2/GCE in 0.1 M PBS with various concentration of (A) AsA (0.1×10^{-6} to 0.5×10^{-3} M) and (B) UA (0.1×10^{-6} to 6.2×10^{-5} M) under f = 10 Hz and E_A = 50 mV.

3.3.1 Voltammetic studies in the mixtures of ascorbic acid and uric acid

The 1wt% Cr-SnO_2 modified GC electrode is capable of distinguishing AsA and UA when they coexist in the same solution. SWV scans (Figure 8) show two distinct peaks with E_p (AsA)= 0.17 V and E_p (UA) = 0.34 V. It can also be noted that the 1wt% Cr-SnO_2 electrode retains its sensitivity to AsA and UA when exposed to their chemical mixtures in various ratios. Figure 8A represents the SWVs at different concentrations of UA where the concentration of AsA was kept constant. The peak current for UA increased linearly with increasing UA concentration with a correlation coefficient of 0.996. The AsA got oxidized at a 170 mV less positive potential than that of UA and the modified electrode can sense even the low concentration level of UA (1 μM) in the presence of a high concentration of AsA (25 μM). The 1wt% Cr-SnO_2/GCE also detects AsA concentrations ranging from 50 μM to 150 μM with a correlation coefficient of 0.995 in the presence of 100 μM UA as shown in Figure 8B.

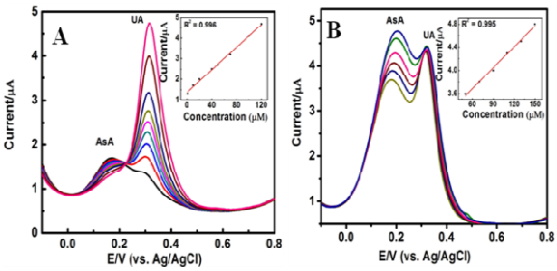

Fig.8 (A) SWVs obtained at 1wt% Cr-SnO_2/GCE in 0.1 M PBS (pH 7.0) (A) containing 25 μM AsA and different concentrations of UA from 1 μM to 120 μM and (B) containing 100 μM UA and different concentrations of AsA from 50 μM to 150 μM. Inset: plots of the anodic peak current as a function of UA and AsA concentrations.

3.4 Anti-interference ability

The electrochemical determination of AsA usually suffer from interferences by dopamine (DA) and uric acid because the oxidation potential for AsA is fairly close to that of the latter bio-molecules. The anti-interference ability of the fabricated sensor was investigated by introducing electroactive species such as 5-fold excess of glucose, sucrose, uric acid, dopamine, Na^{2+}, K^{+} and Mg^{2+}. The influences of the above species on the detection of ascorbic acid at the 1wt% Cr-SnO_2 modified GC electrodes are shown in Figure 9A, which clearly indicates that the investigated interfering species did not have any influence on the determination of ascorbic acid. However, the modified electrode gives current response to 5 fold excess of DA. In order to study the effect of DA, the SWV response of AsA at the 1wt% Cr-SnO_2/GCE was investigated in the presence of 1mM DA. Figure 9B depicts the SWVs of 1wt% Cr-SnO_2/GCE in 10 μM AsA and 50 μM UA mixture (curve a) and a mixture of 1 mM DA, 10 μM AsA and 50 μM (curve b). Compared with AsA and UA alone, Epa for AsA in the presence of DA is positively shifted by about 20 mV. Nevertheless this will not affect the determination of AsA by voltammetry method, as the addition of DA enhances the peak current value of AsA. Thus, it is clearly seen that the excess DA does not affect the catalytic oxidation of AsA. The result suggests that the fabricated electrodes are favorable for the selective determination of ascorbic acid in the presence of common interfering species including DA.

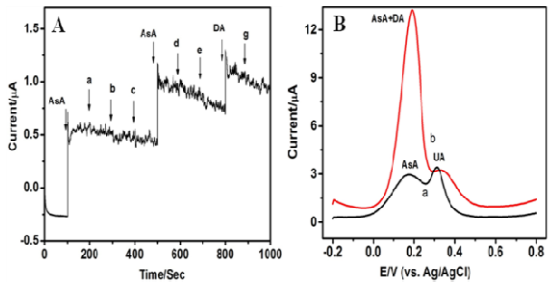

Fig. 9 (A) Amperometric response of the nano Cr-SnO_2 modified electrode for the addition of 5 μM AsA with successive additions of interferences (25 μM) in the sequence of (a) glucose, (b) sucrose, (c) uric acid, (d) NaCl, (e) KCl and (f) DA and (g) $MgCl_2$ in 0.1 M PB solution. (B) SWVs of Cr-SnO_2/GCE in the presence of 10 μM AsA and 50 μM UA (a), 1 mM DA, 10 μM AsA and 50 μM UA (b).

3.5 Reproducibility and stability

To evaluate the reproducibility of 1wt% Cr-SnO_2/GCE, the peak currents of 5 successive measurements by CV in a mixture solution of 500 μM AsA and UA were determined. The relative standard deviation (R.S.D.) of 1.5% and 1.2% was obtained for AsA and UA respectively, indicating that the nano Cr- SnO_2/GCE is not subject to surface fouling by the oxidation products. The stability of the sensor was investigated by recording cyclic voltammograms of 1wt% Cr-SnO_2/GCE in 100 μM ascorbic acid and uric acid. After 25 successive cycles, the anodic current of modified electrode decreased only by 3.5 % and 2.8 % for AsA and UA respectively. The above results showed that the present modified electrode was very much stable and reproducible towards AsA and UA determination.

3.6 Determination of AsA and UA in pharmaceutical products, fruits and human urine, blood samples

The applicability of the 1wt% Cr-SnO_2/GCE for amperomertric determination of AsA was verified through analysis of real samples such as pharmaceutical products and fruits. B complex forte with vitamin C, B complex with vitamin C and biotin and vitamin C tablet with the AsA declared content of 150, 150, 500 mg/tablet respectively were chosen for the investigations. Solutions were prepared as described in Section 2.3. Every analysis was repeated for 3 times and the results (Table 2) are consistent with the contents declared by producers in the analyzed vitamin preparations. The quantity of AsA observed in various fruit samples are shown in Table 3. It is, therefore, possible to conclude that 1wt% Cr-SnO_2/GCE can serve as

suitable tool for AsA determination in pharmaceutical products and fruits.

The sensor was also applied to determine the concentration of UA in human urine and blood samples. The samples were diluted to 20 times with 0.1 M PBS (pH 7.0) without any further treatment. Then, 100 μL diluted urine sample was added into a small beaker containing 20 mL PBS for electrochemical determination. In addition, different concentrations of UA standard solution are added for testing recoveries with three repetitions. And the data were analyzed by applying a calibration plot obtained from the quantitative determination of UA. The results were shown in Table 3. The RSD was acceptable, revealing that the proposed method could be efficiently used for the determination of UA in human urine and blood samples.

TABLE 2 DETERMINATION OF ASA IN PHARMACEUTICAL PRODUCTS AND FRUITS

Samples	Labelled content (mg/tablet)	Observed content (mg/tablet)	RSD (n=3) (%)
B complex Forte with vitamin C	150	148.1±0.07	2.07
B complex with vitamin C-biotin	150	141.5±0.12	0.71
Vitamin C Tablet	500	443.2±0.05	0.45
Orange	-	7.04 ± 0.06	0.59
Pomegranate	-	5.63±0.15	0.74
Tomato	-	4.91±0.03	0.91
Lemon	-	3.52±0.13	2.75

TABLE 3 DETERMINATION OF UA IN URINE AND SERUM USING 1WT% CR-SnO_2/GCE

Sample	Added (μM)	Found (μM)	RSD (n=3) (%)
Urine	3	6.4	4.72
Urine	5	9.2	1.94
Blood Serum	5	7.8	1.57
Blood Serum	5	8.6	2.38

IV. CONCLUSION

We have successfully fabricated a sensor for simultaneous detection of ascorbic acid and uric acid in the presence of dopamine using Cr - SnO_2 NPs as sensing element. The 1wt% Cr-SnO_2 NPs modified glassy carbon electrode exhibited the lowest detection limit of 36 nM and 14 nM over a wide linear range for AsA (0.1×10^{-6} to 0.5×10^{-3} M) and UA (0.1×10^{-6} to 6.2×10^{-5} M) respectively. The fabricated sensor was proved to be useful for the quantification of ascorbic acid and uric acid in pharmaceutical samples, fruits and human urine, blood serum samples. The advantageous features such as easy fabrication and low cost can make the proposed sensor attractive for applications in health care and food industries.

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